

College of Engineering and Technology

Project Work/Internship – Student Hand Book

Project Batch ID
752

Degree/ program	B.Tech	Specialisation	Computer Science and Engineering
Academic Year	2025-2026 (Even)	Semester	8

Name of student	Register Number	Department	Mobile Number	Email ID
Shovik Banerjee	RA2211003010270	C.Tech	9007837763	Sb6935@srmist.edu.in
Kaushani Chakraborty	RA2211003010227	C.Tech	8420605161	Kc7009@srmist.edu.in

Working Title of the Project:	Unified Transformer-Based Spatio-Temporal Forecasting of Air Quality and Urban Noise Risk with a Rule-Based Public Advisory Layer
Project Site / Location	Chennai
Name and address of the company/organisation (Applicable for projects with industry or industry support)	SRM IST, Kattankulathur, Chengalpattu District-603203

Supervision Team

	Supervisor	Co-Supervisor	External Supervisor (If applicable)
Name	Dr Akella S Narasimha Raju		
Designation	Assistant Professor		
Department	C.Tech		
Campus	Kattankulathur		
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E-mail	ar9488@srmist.edu.in		

College of Engineering and Technology

Project Work/Internship – Student Hand Book

Course Code	21CSP401L/ 21CSP402L	Course Title	Project
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Mission Statement

Problem (or) Product Description:

Rapid urbanization has led to a significant increase in air and noise pollution, posing serious risks to public health and quality of life. Existing systems primarily provide isolated, static pollution measurements without predictive capability or source identification. There is a need for an integrated, real-time system that can accurately predict air and noise pollution levels and identify their probable sources to support informed urban planning and mitigation strategies

Assumptions and Constraints

1. The system assumes the availability of reliable historical air and noise pollution data for model training.
2. Predictions are constrained to locations covered by existing monitoring stations.
3. Pollution source identification is inferred from learned patterns, not direct measurements.
4. Model accuracy may be affected by sudden events or extreme weather conditions.
5. Real-time performance depends on computational resources and network latency.

Stakeholders

- Urban residents and commuters affected by air and noise pollution.
- Government and municipal authorities responsible for environmental monitoring and regulation.
- Urban planners and policymakers using data to design mitigation strategies.
- Environmental agencies and researchers analyzing pollution trends and impacts.
- Healthcare organizations concerned with pollution-related public health outcomes.

College of Engineering and Technology

Project Work/Internship – Student Hand Book

Division of work and contributors

Time period		Activities or components of the project	Name/Register Number of the Individual Contributor	Names/Register Number of the Joint Contributors
From Date	To Date			
08/11/25	13/11/25	Literature survey and identification of research gaps in Air and noise Pollution	Shovik Banerjee (RA2211003010270)	Kaushani Chakraborty (RA2211003010227)
3/12/25	6/12/25	Collection and examination of open- access air and noise datasets.	Kaushani Chakraborty (RA2211003010227)	Shovik Banerjee (RA2211003010270)
7/12/25	10/12/25	Data preprocessing, temporal alignment, and feature selection.	Kaushani Chakraborty (RA2211003010227)	Shovik Banerjee (RA2211003010270)
11/12/25	15/12/25	Time-series construction and preparation of training, validation, and test datasets.	Shovik Banerjee (RA2211003010270)	Kaushani Chakraborty (RA2211003010227)
15/12/25	17/12/25	Training of model on the processed Dataset	Shovik Banerjee (RA2211003010270)	Kaushani Chakraborty (RA2211003010227)
17/12/25	23/12/25	Hyper tuning the model to improve accuracy	Kaushani Chakraborty (RA2211003010227)	Shovik Banerjee (RA2211003010270)
31/12/25	15/01/26	Comparative analysis of the result	Shovik Banerjee (RA2211003010270)	Kaushani Chakraborty (RA2211003010227)
18/01/26	20/02/26	Journal writing and submission	Shovik Banerjee (RA2211003010270)	Kaushani Chakraborty (RA2211003010227)

College of Engineering and Technology

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College of Engineering and Technology

Project Work/Internship – Student Hand Book

Summary record of major progress meetings with supervisors (Sprint 1)

Summary record of major progress meetings with supervisors			Working title of dissertation/research project:	
Meeting date & supervisors present	Progress since last meeting	Agreed programme of work and target dates	Other issues, e.g. facilities, supervision, training needs, etc.	Date of next meeting
10/11/2025	Identified Research gaps. Collected and Examined open-access Air and Noise dataset. Trained and evaluated the model Performance	Before next review 50% of project completed		5/1/26
06/01/2026	Paper Draft written	Paper Submission	Supervisor Guidance	07/03/26
07/03/2026	Paper submission Paper accepted		Supervisor Guidance	

College of Engineering and Technology

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Summary record of major progress meetings with supervisors (SPRINT 2)

Summary record of major progress meetings with supervisors			Working title of dissertation/research project:	
Meeting date & supervisors present	Progress since last meeting	Agreed programme of work and target dates	Other issues, e.g. facilities, supervision, training needs, etc.	Date of next meeting

College of Engineering and Technology

Project Work/Internship – Student Hand Book

Summary record of major progress meetings with supervisors (SPRINT 3)

Summary record of major progress meetings with supervisors			Working title of dissertation/research project:	
Meeting date & supervisors present	Progress since last meeting	Agreed programme of work and target dates	Other issues, e.g. facilities, supervision, training needs, etc.	Date of next meeting

College of Engineering and Technology

Project Work/Internship – Student Hand Book

Worksheet / Data collection / Observation etc

Data collection

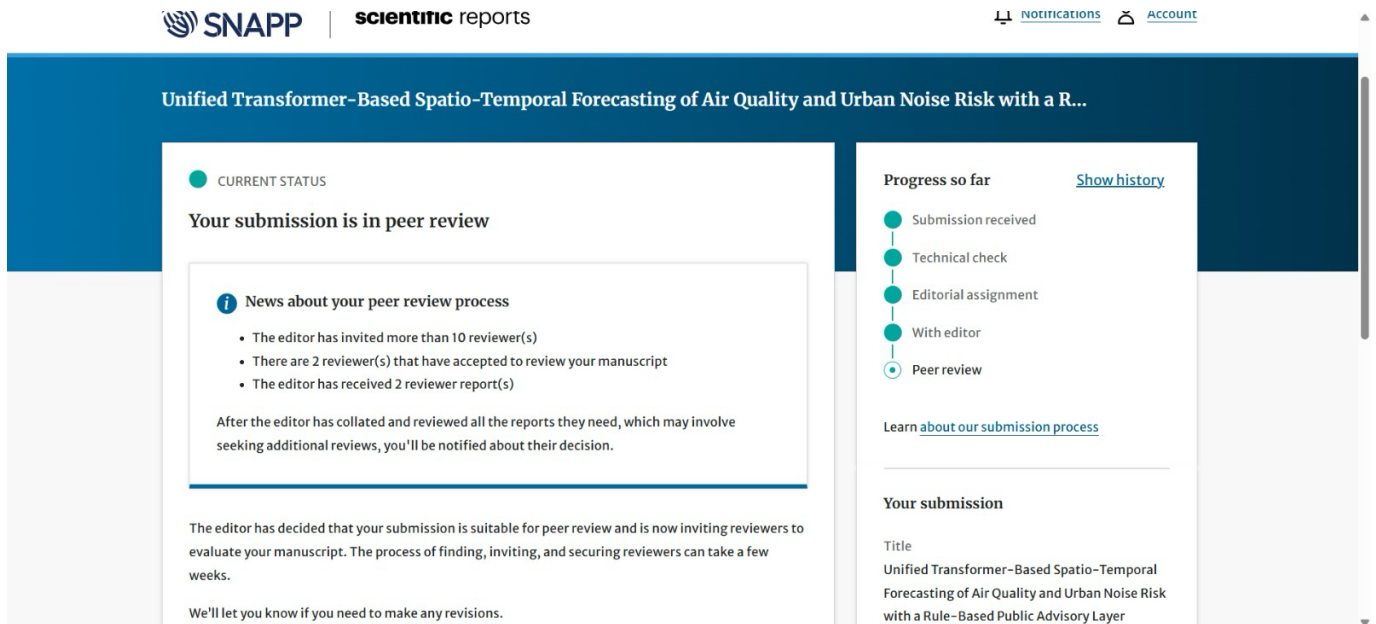
Air pollution dataset:- <https://www.kaggle.com/datasets/rohanrao/air-quality-data-in-india>

Noise pollution dataset:- <https://www.kaggle.com/datasets/rohanrao/noise-monitoring-data-in-india>

College of Engineering and Technology

Project Work/Internship – Student Hand Book

Journal Publication Details



SNAPP | scientific reports

Unified Transformer-Based Spatio-Temporal Forecasting of Air Quality and Urban Noise Risk with a R...

CURRENT STATUS

Your submission is in peer review

News about your peer review process

- The editor has invited more than 10 reviewer(s)
- There are 2 reviewer(s) that have accepted to review your manuscript
- The editor has received 2 reviewer report(s)

After the editor has collated and reviewed all the reports they need, which may involve seeking additional reviews, you'll be notified about their decision.

The editor has decided that your submission is suitable for peer review and is now inviting reviewers to evaluate your manuscript. The process of finding, inviting, and securing reviewers can take a few weeks.

We'll let you know if you need to make any revisions.

Progress so far [Show history](#)

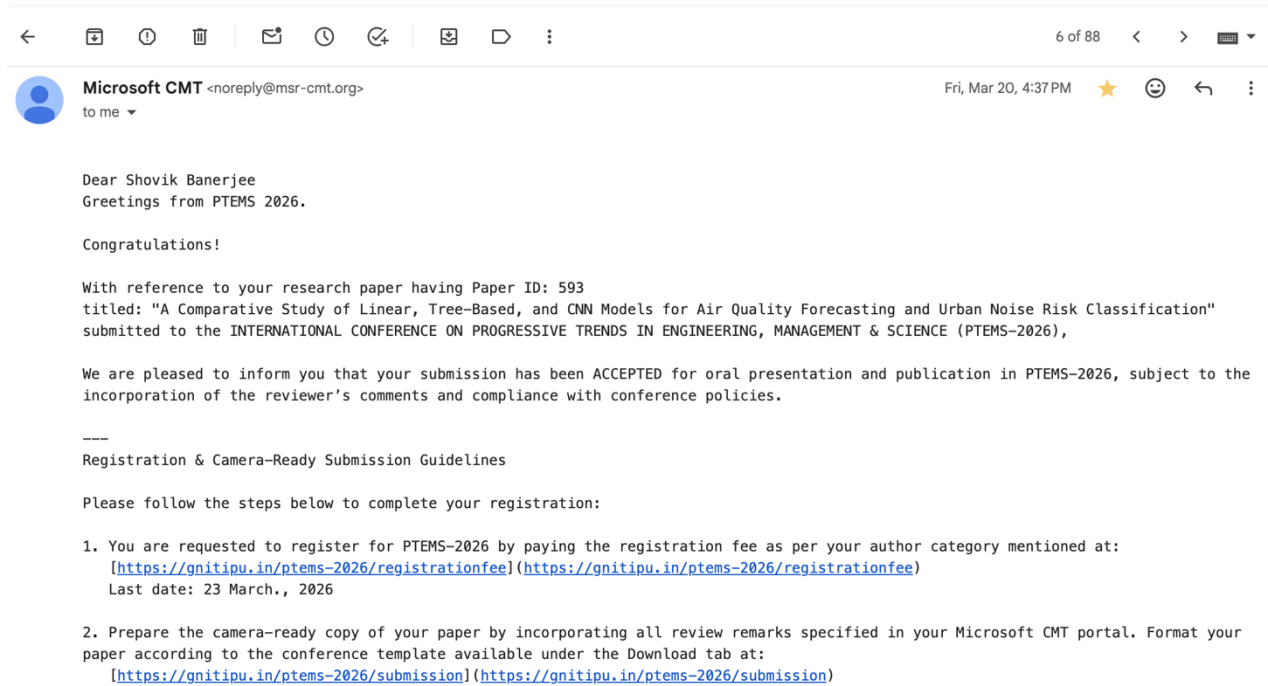
- Submission received
- Technical check
- Editorial assignment
- With editor
- Peer review**

[Learn about our submission process](#)

Your submission

Title
Unified Transformer-Based Spatio-Temporal Forecasting of Air Quality and Urban Noise Risk with a Rule-Based Public Advisory Layer

Conference Acceptance



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Microsoft CMT <noreply@msr-cmt.org> Fri, Mar 20, 4:37 PM ★ 😊 ↶ ⋮
to me ▾

Dear Shovik Banerjee
Greetings from PTEMS 2026.

Congratulations!

With reference to your research paper having Paper ID: 593
titled: "A Comparative Study of Linear, Tree-Based, and CNN Models for Air Quality Forecasting and Urban Noise Risk Classification"
submitted to the INTERNATIONAL CONFERENCE ON PROGRESSIVE TRENDS IN ENGINEERING, MANAGEMENT & SCIENCE (PTEMS-2026),

We are pleased to inform you that your submission has been ACCEPTED for oral presentation and publication in PTEMS-2026, subject to the incorporation of the reviewer's comments and compliance with conference policies.

Registration & Camera-Ready Submission Guidelines

Please follow the steps below to complete your registration:

- You are requested to register for PTEMS-2026 by paying the registration fee as per your author category mentioned at:
<https://gnitipu.in/ptems-2026/registrationfee> (<https://gnitipu.in/ptems-2026/registrationfee>)
Last date: 23 March., 2026
- Prepare the camera-ready copy of your paper by incorporating all review remarks specified in your Microsoft CMT portal. Format your paper according to the conference template available under the Download tab at:
<https://gnitipu.in/ptems-2026/submission> (<https://gnitipu.in/ptems-2026/submission>)

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Project Work/Internship – Student Hand Book

ICPTMS/P/2026/ 127



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CERTIFICATE

OF PRESENTATION

This is to certify that Dr./Mr./Ms. Shovik Banerjee affiliated to SRM Institute of Science and Technology, Kattankulathur, Chennai, Tamilnadu, India has presented the paper titled A Comparative Study of Linear, Tree-Based, and CNN Models for Air Quality Forecasting and Urban Noise Risk Classification in the International Conference On Progressive Trends In Engineering, Management & Science (PTEMS-2026) Held On 10-11 April-2026 Organized By Greater Noida Institute of Technology (GNIT-IPU Campus), Greater Noida, Uttar Pradesh, India.



Prof. Sujeet Kumar
Conference Organising Chair



Prof. (Dr.) Somendra Shukla
Conference General Chair

Presentation Certificat

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Project Work/Internship – Student Hand Book

